MATH 8090:
Spectral Analysis of Time Series II

Readings: CC08 Chapter 14; BD16 Chapters 4 and 10.1; SS17 Chapters 1.5–1.6, 4.4–4.6, 4.8, and 5.5

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Week 12

Agenda

Review

Southern Oscillation Index Example

Parametric Spectral Estimation

Lagged Regression Models

Review: Nonparametric Spectral Estimation

- Periodogram: $I(\omega_j) = |d(\omega_j)|^2$, where

$$d(\omega_j) = n^{-\frac{1}{2}} \sum_{t=1}^{n} y_t e^{-2\pi i \omega_j t}, \ \omega_j = \frac{j}{n}, \ j = 0, 1, \cdots, n-1$$

  - $\frac{I(\omega_j)}{\frac{1}{2} f(\omega_j)} \overset{\text{approx.}}{\sim} \chi^2_2$, $j = 1, \cdots, m = \frac{n-1}{2} \Rightarrow \mathbb{E}[I(\omega_j)] \approx f(\omega_j)$ (unbiased)
  - But $\text{Var}[I(\omega_j)] \approx f^2(\omega_j)$ (inconsistent because it does not go to 0 as $n \to \infty$)

- Smooth the periodogram
  - Averaged periodogram: $\bar{f}(\omega_j) = \frac{1}{L} \sum_{k=-l}^{l} I(\omega_{j+k}), \quad L = 2l + 1$
  - Smoothed periodogram: $\bar{f}(\omega_j) = \sum_{k=-l}^{l} W_l(k) I(\omega_{j+k})$

- Pointwise CI for $f(\omega_j)$:

$$\frac{\nu \bar{f}(\omega_j)}{\chi^2_\nu(1-\alpha/2)} \leq f(\omega_j) \leq \frac{\nu \bar{f}(\omega_j)}{\chi^2_\nu(\alpha/2)}, \quad \nu = \frac{2}{\sum_{k=-l}^{l} W_l^2(k)}$$

Spectral ANOVA

- For odd $n = 2m + 1$, the inverse transform can be written

$$y_t - \bar{y} = \frac{2}{\sqrt{n}} \sum_{j=1}^{m} \left[d_{\cos}(\omega_j) \cos(2\pi\omega_j t) + d_{\sin}(\omega_j) \sin(2\pi\omega_j t) \right].$$

- Square and sum over t; orthogonality of sines and cosines implies that

$$\begin{aligned} \sum_{t=1}^{n} (y_t - \bar{y})^2 &= 2 \sum_{j=1}^{m} \left[d_{\cos}(\omega_j)^2 + d_{\sin}(\omega_j)^2 \right] \\ &= 2 \sum_{j=1}^{m} I(\omega_j) \end{aligned}$$

We have partitioned $\sum_{t=1}^{n} (y_t - \bar{y})^2$ into $2 \times \sum_{j=1}^{m} I(\omega_j)$. This leads to Spectral ANOVA $\Rightarrow$ total variance = sum of frequency components

Spectral ANOVA (Cont'd)

Source	df	SS	MS
ω_1	2	$2I(\omega_1)$	$I(\omega_1)$
ω_2	2	$2I(\omega_2)$	$I(\omega_2)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
ω_m	2	$2I(\omega_m)$	$I(\omega_m)$
Total	$2m = n - 1$	$\sum(y_t - \bar{y})^2$	

Toy example:

```
> x <- c(1, 2, 3, 2, 1) - mean(x)
> c1 <- cos(2 * pi * (1:5) * (1 / 5)); s1 <- sin(2 * pi * (1:5) * (1 / 5))
> c2 <- cos(2 * pi * (1:5) * (2 / 5)); s2 <- sin(2 * pi * (1:5) * (2 / 5))
> omega1 <- cbind(c1, s1); omega2 <- cbind(c2, s2)
> anova(lm(x ~ omega1 + omega2))
Warning in anova.lm(lm(x ~ omega1 + omega2)) :
  ANOVA F-tests on an essentially perfect fit are unreliable
Analysis of Variance Table

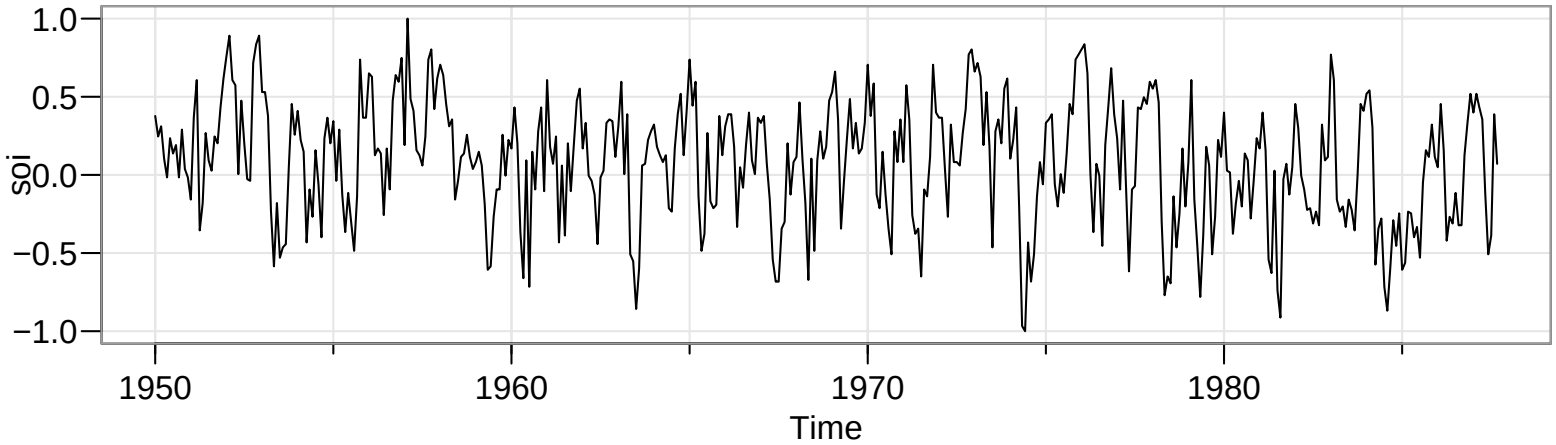
Response: x
          Df  Sum Sq Mean Sq F value Pr(>F)
omega1     2 2.74164 1.37082
omega2     2 0.05836 0.02918
Residuals  0 0.00000
```

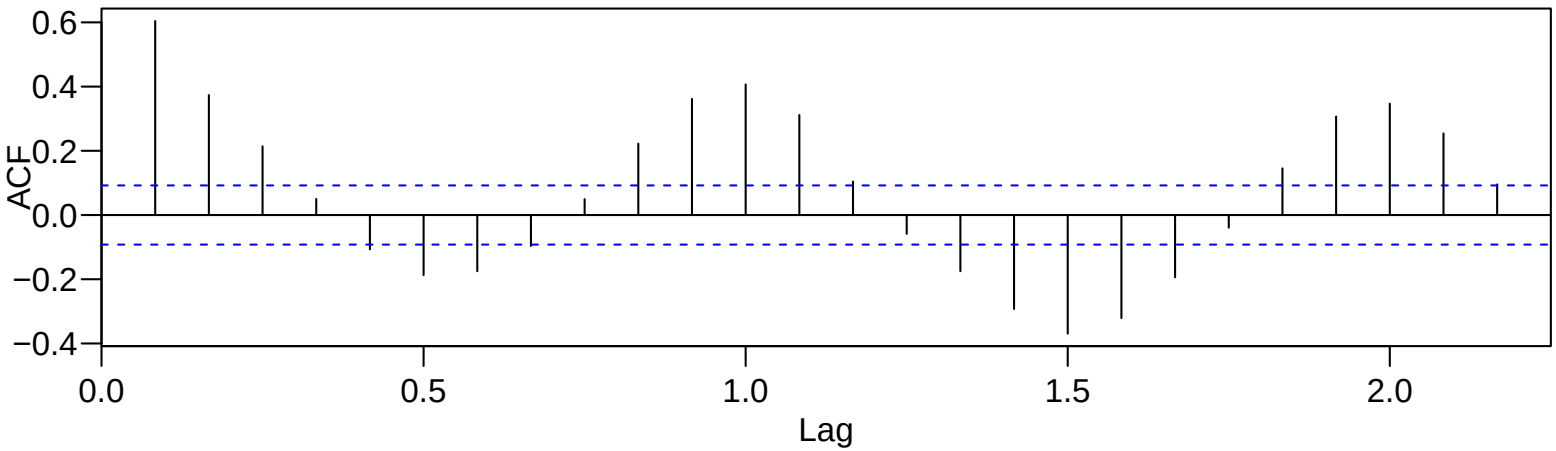
This Week: Three Questions

1. Which frequencies dominate a real time series? SOI and spectral ANOVA
2. Can a fitted ARMA model estimate its spectrum? Parametric spectra
3. How are two series related across lags and frequencies? CCF and coherence

Example: Southern Oscillation Index (SOI)

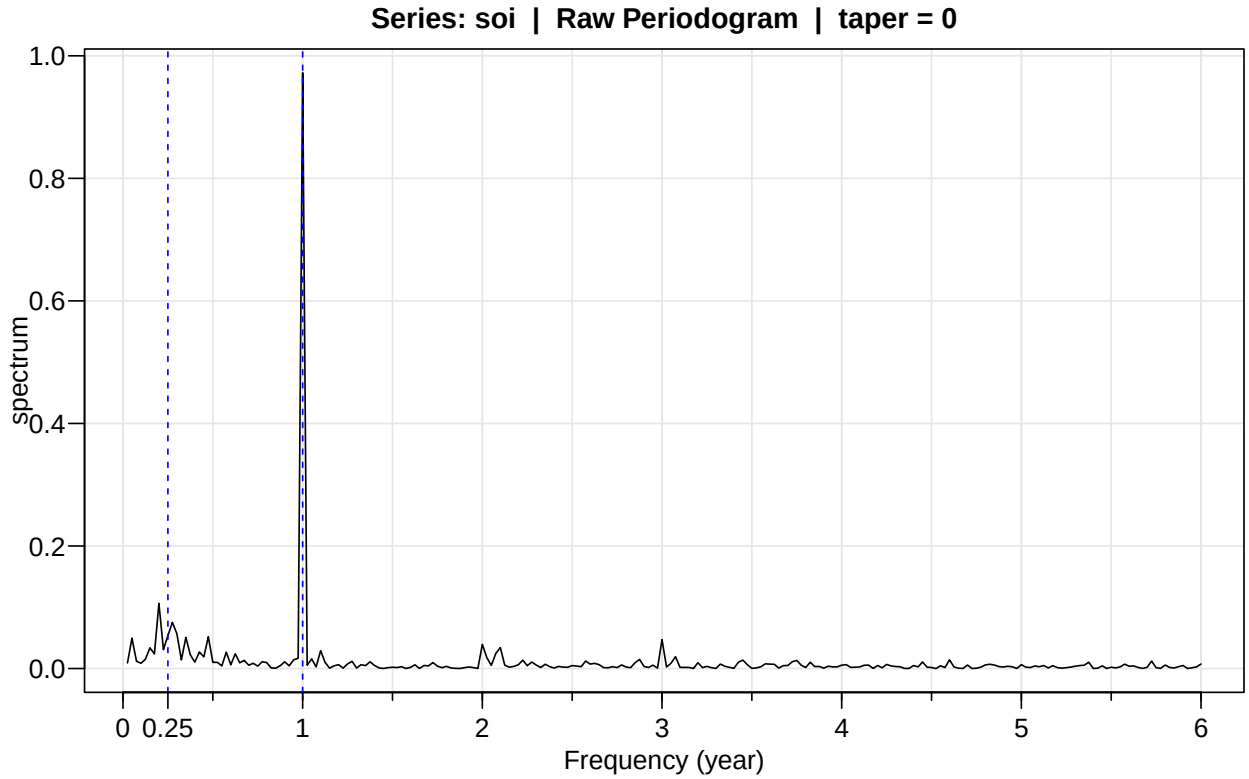
Southern Oscillation Index (SOI) for a period of 453 months ranging over the years 1950-1987





What are the hidden periods of SOI?

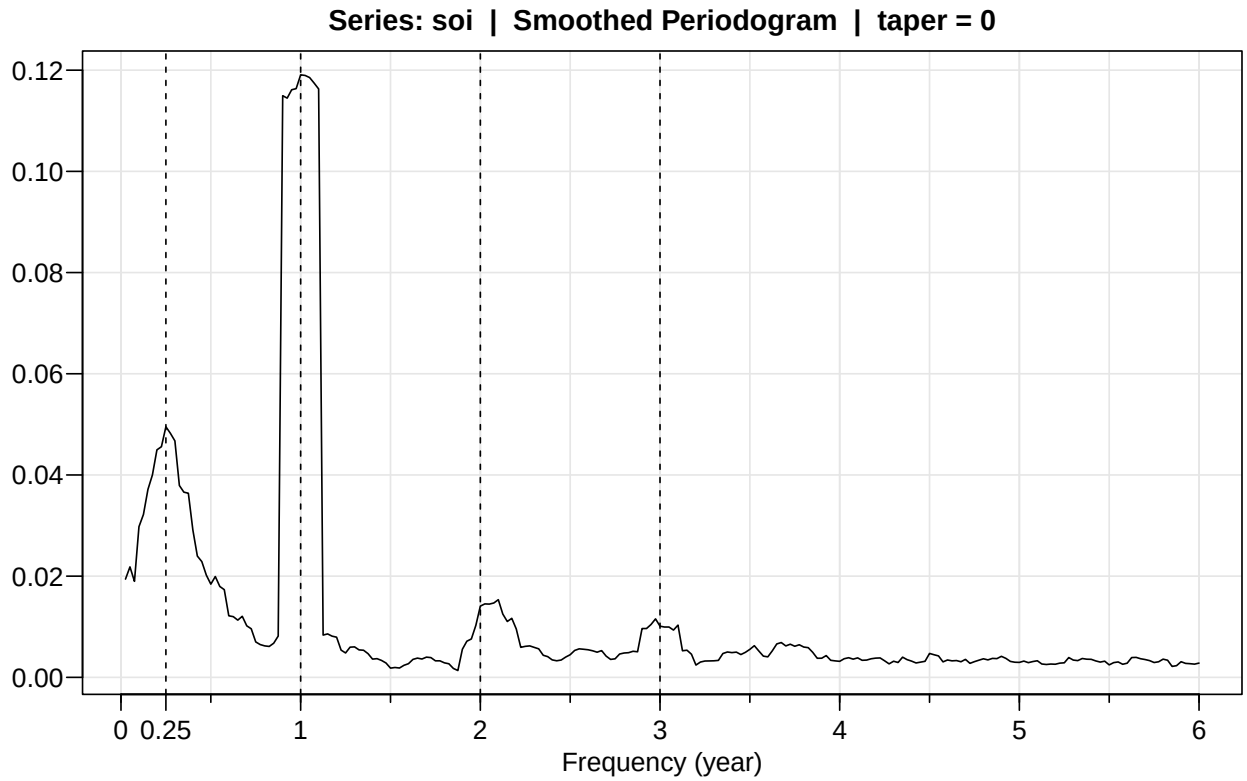
SOI Example: Raw Periodogram



An approximate 95% confidence interval for $f(\omega)$:

ω	Period	Power	Lower	Upper
$\frac{1}{48}$	4 years	0.0537	0.0146	2.1222
$\frac{1}{12}$	1 year	0.9722	0.2636	38.4011

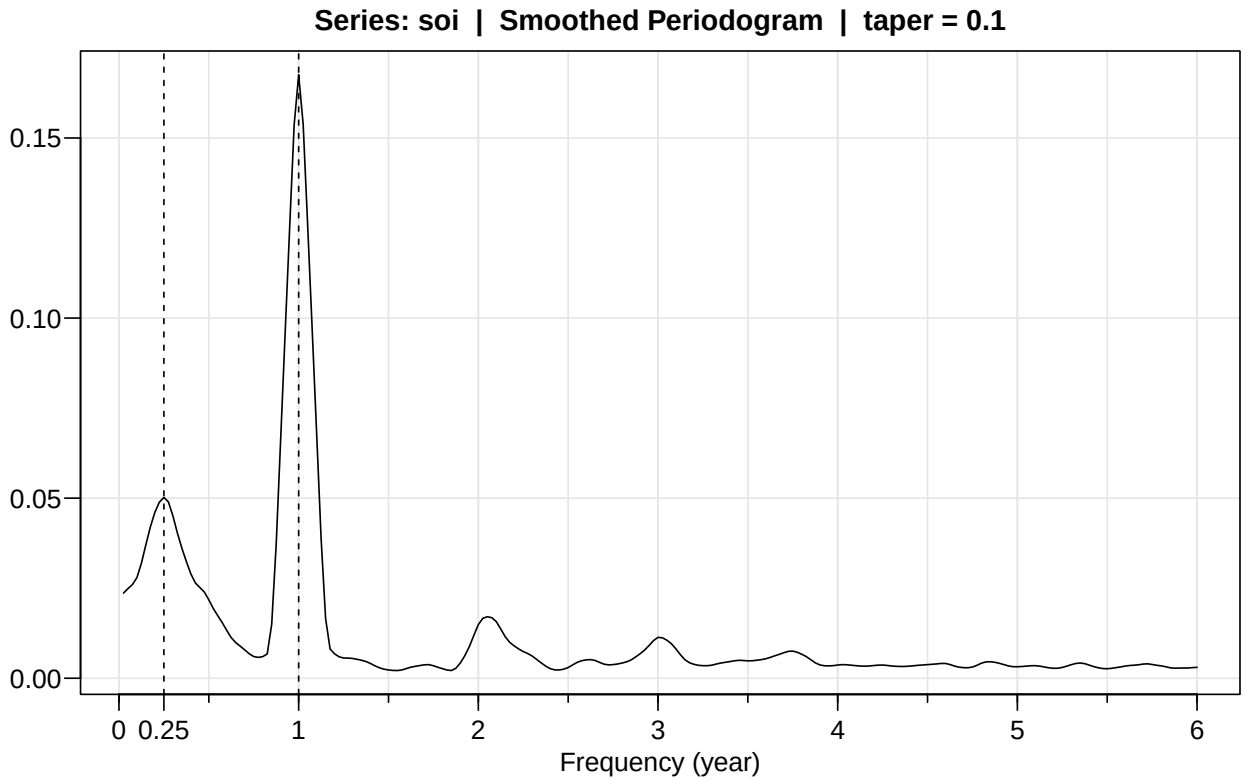
SOI Example: Averaged Periodogram (Daniell with $l = 4$)



An approximate 95% confidence interval for $f(\omega)$:

ω	Period	Power	Lower	Upper
$\frac{1}{48}$	4 years	0.0495	0.0279	0.1113
$\frac{1}{12}$	1 year	0.1191	0.0670	0.2677

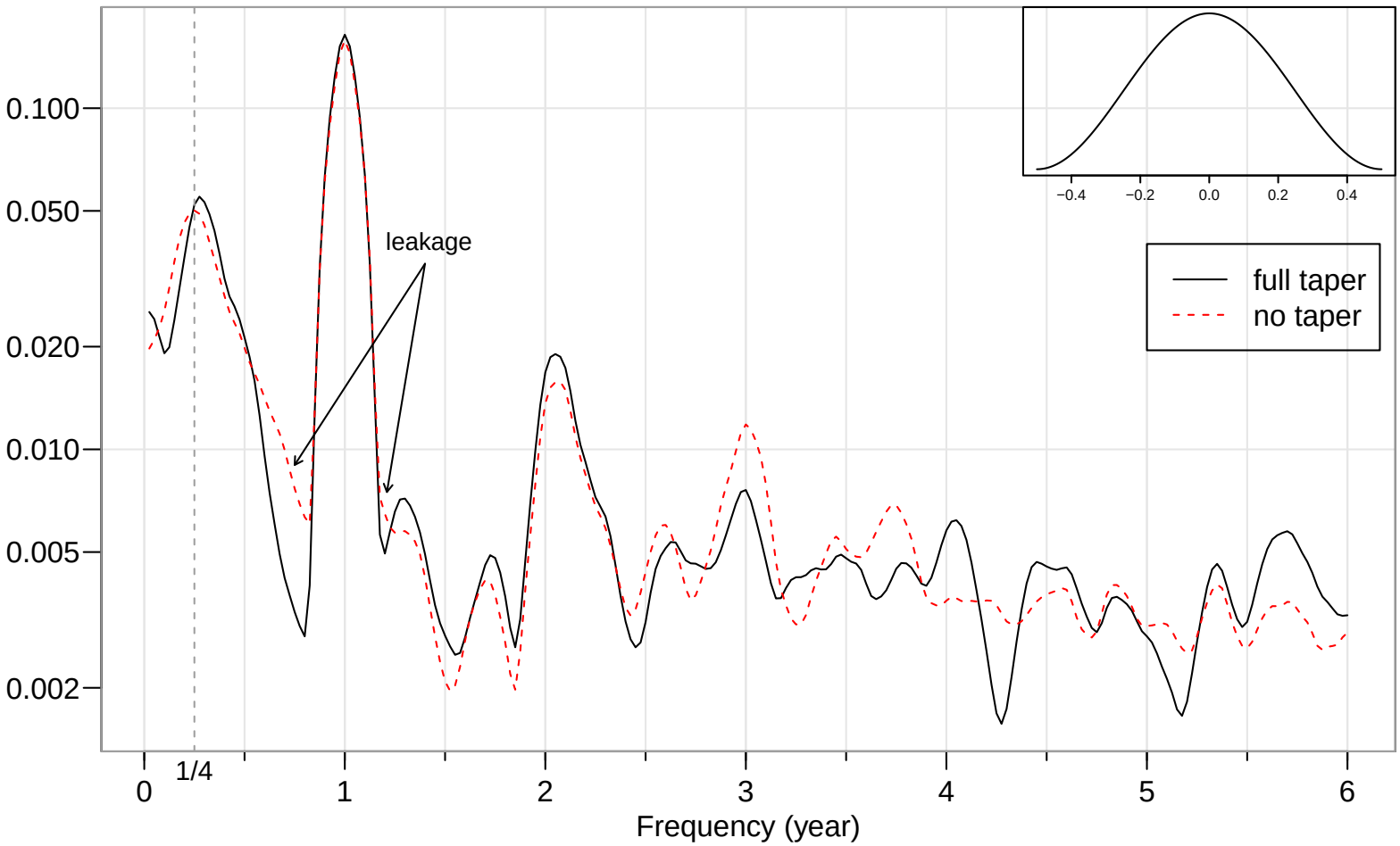
SOI Example: Smoothed Periodogram (modified Daniell $c(3,3)$)



An approximate 95% confidence interval for $f(\omega)$:

ω	Period	Power	Lower	Upper
$\frac{1}{48}$	4 years	0.0502	0.0283	0.1129
$\frac{1}{12}$	1 year	0.1675	0.0943	0.3767

Apply Tapering to Alleviate Spectral Leakage



The tapered spectrum does a better job in separating the annual cycle (period 12 months) and a roughly four-year cycle (period 48 months)

Seasonally Adjusted SOI [**Source**: Peter Bloomfield's ST 730 Lecture Notes]

- The Southern Oscillation Index data provided by Shumway and Stoffer is not seasonally adjusted, which explains the substantial peaks in the periodogram at the annual frequency

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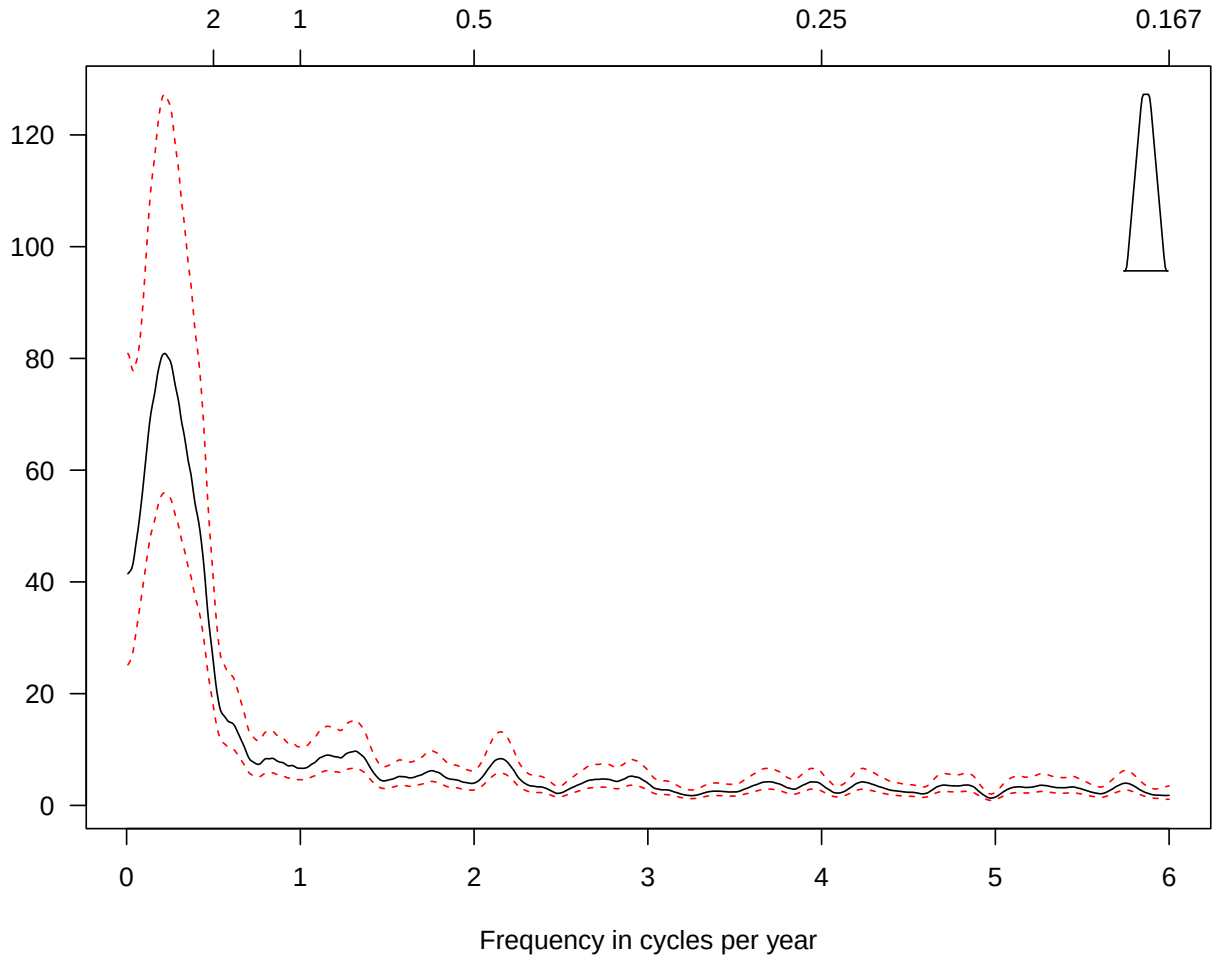
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- A deterministic seasonal mean violates weak stationarity; after removing it, we can study the spectrum of the remaining component if that component is stationary

Seasonally Adjusted SOI [**Source**: Peter Bloomfield's ST 730 Lecture Notes]

- The Southern Oscillation Index data provided by Shumway and Stoffer is not seasonally adjusted, which explains the substantial peaks in the periodogram at the annual frequency
- A deterministic seasonal mean violates weak stationarity; after removing it, we can study the spectrum of the remaining component if that component is stationary
- A more sensible analysis uses the seasonally adjusted series. (Bloomfield did this by fitting a seasonal means model using data from 1876-2010.)

SOI Example from Bloomfield: Smoothed Periodogram



Note that the peak at the annual frequency disappears due to the removal of the annual cycle

Interpreting the SOI Spectral Comparisons

- Raw periodogram: high resolution, but very high variability.
- Daniell smoothing: averages nearby frequencies; peaks become more stable.
- Tapering: mitigates leakage from strong components.
- Seasonal adjustment: removes a deterministic seasonal mean, if present.

A peak near an annual frequency does not, by itself, establish nonstationarity: stationary processes can also exhibit seasonal spectral peaks.

Parametric versus Nonparametric Estimation

- Parametric estimation: estimate a model that is specified by a fixed number of parameters

- Nonparametric estimation: estimate a model that is specified by a number of parameters that can grow as the sample grows

The smoothed periodogram estimates we have considered are **nonparametric**: the estimates of the spectral density can be parameterized by estimated values at ω_j's. As $n \uparrow$, the number of distinct frequency values increases

The time domain models we considered are **parametric**. For example, an ARMA(p,q) process can be completely specified with $p + q + 1$ parameters

Parametric Spectral Estimation

The typical approach is to use the maximum likelihood parameter estimates $(\hat{\phi}_1, \cdots, \hat{\phi}_p, \hat{\sigma}^2)$ for the parameters of an AR(p), and then compute $f(\omega)$ for this estimated AR model:

$$\hat{f}(\omega) = \frac{\hat{\sigma}^2}{|\hat{\phi}(e^{-2\pi i\omega})|^2}$$

For large n,

$\mathrm{Var}(\hat{f}(\omega))$ depends on model, frequency, and estimation method

- Increasing p allows more complex spectral shapes and may reduce approximation bias

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- Increasing p allows more complex spectral shapes and may reduce approximation bias
- Larger p typically increases estimation uncertainty and overfitting risk

ARMA Spectral Estimation

- Sometimes ARMA models are used instead
- Estimate the parameters of an ARMA(p,q) model and compute its spectral density:

$$\hat{f}(\omega) = \hat{\sigma}^2 \left| \frac{\hat{\theta}(e^{-2\pi i\omega})}{\hat{\phi}(e^{-2\pi i\omega})} \right|^2 .$$

- However, it is more common to use large AR models, rather than ARMA models

Parametric versus Nonparametric Spectral Estimation

- The main advantage of parametric spectral estimation over nonparametric is that it often gives better frequency resolution of a small number of peaks

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- The main advantage of parametric spectral estimation over nonparametric is that it often gives better frequency resolution of a small number of peaks
- This is especially important if there is more than one peak at nearby frequencies
- The disadvantage of parametric spectral estimation is the inflexibility due to the use of the restricted class of ARMA models.

Parametric Spectral Estimation: Summary

Given data $y_1, y_2, \cdots, y_n$,

1. Estimate the AR parameters $(\phi_1, \phi_2, \cdots, \phi_p, \sigma^2)$ using maximum likelihood or Yule-Walker/least squares, choose a suitable model order p using AIC or BIC

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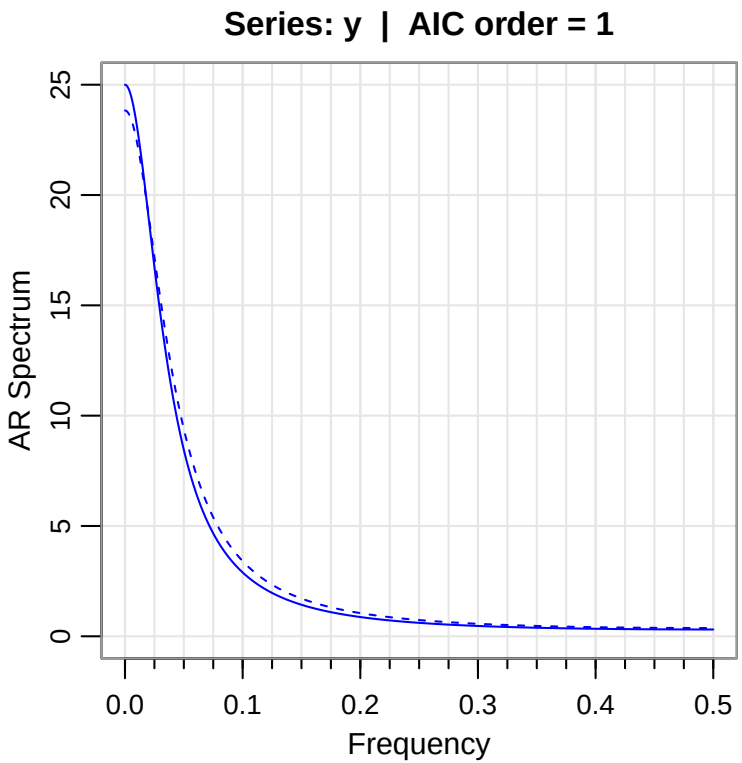
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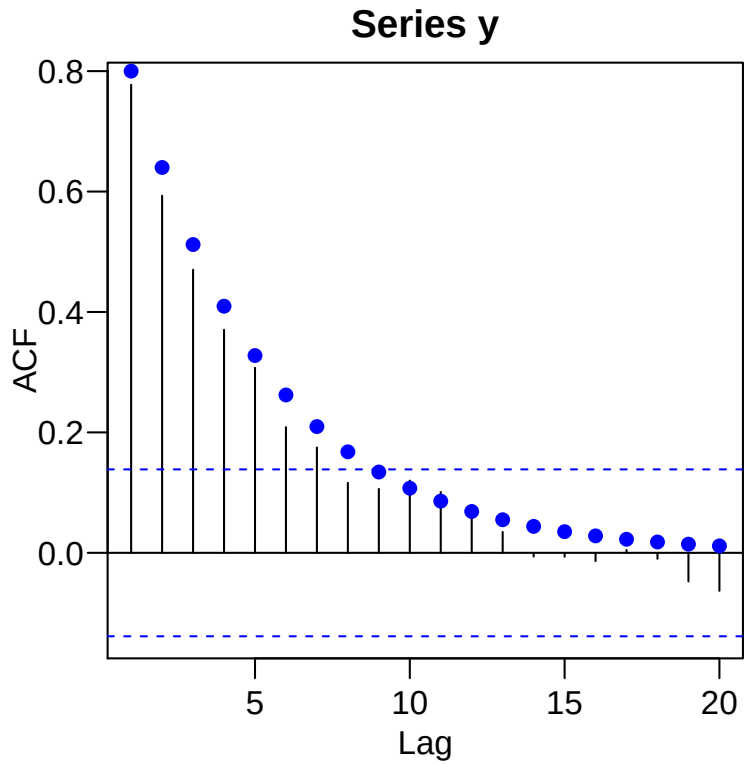
2. Use the estimates $(\hat{\phi}_1, \hat{\phi}_2, \cdots, \hat{\phi}_p, \hat{\sigma}^2)$ to compute the estimated spectral density:

$$\hat{f}(\omega) = \frac{\hat{\sigma}^2}{\left|\hat{\phi}(e^{-2\pi i\omega})\right|^2}$$

Example: AR(1) with $\phi = 0.8$

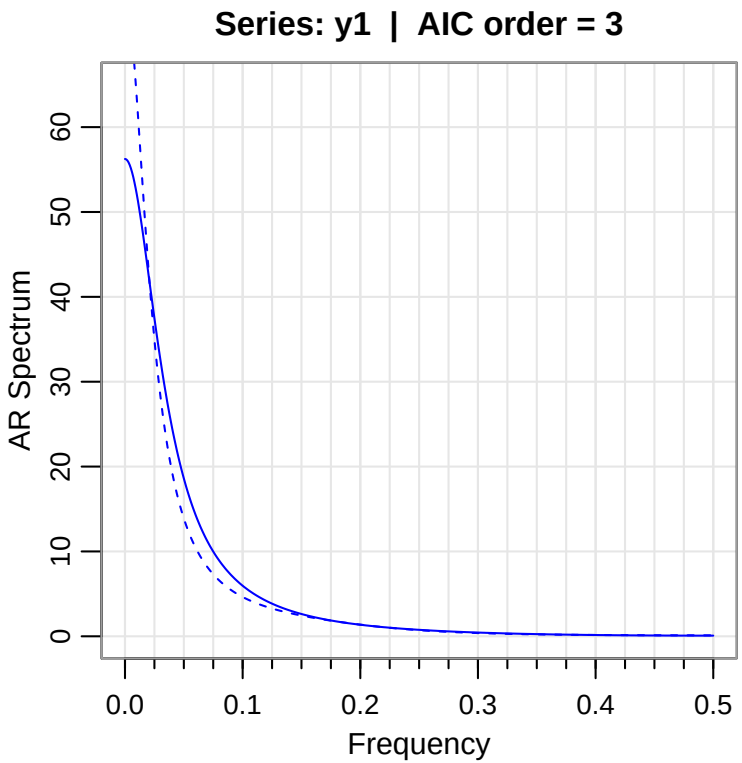
1. Use AIC to select p, the order of the AR model
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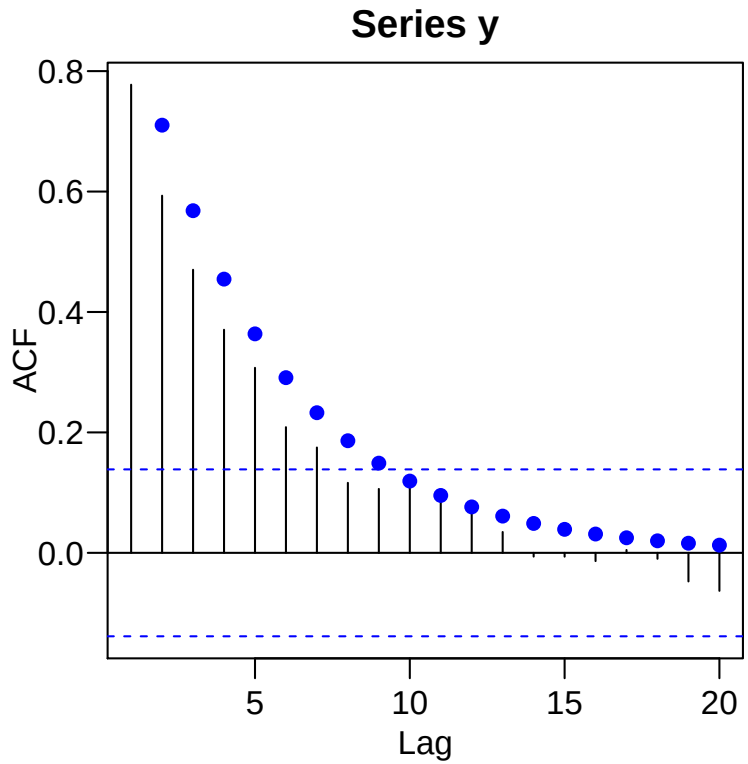




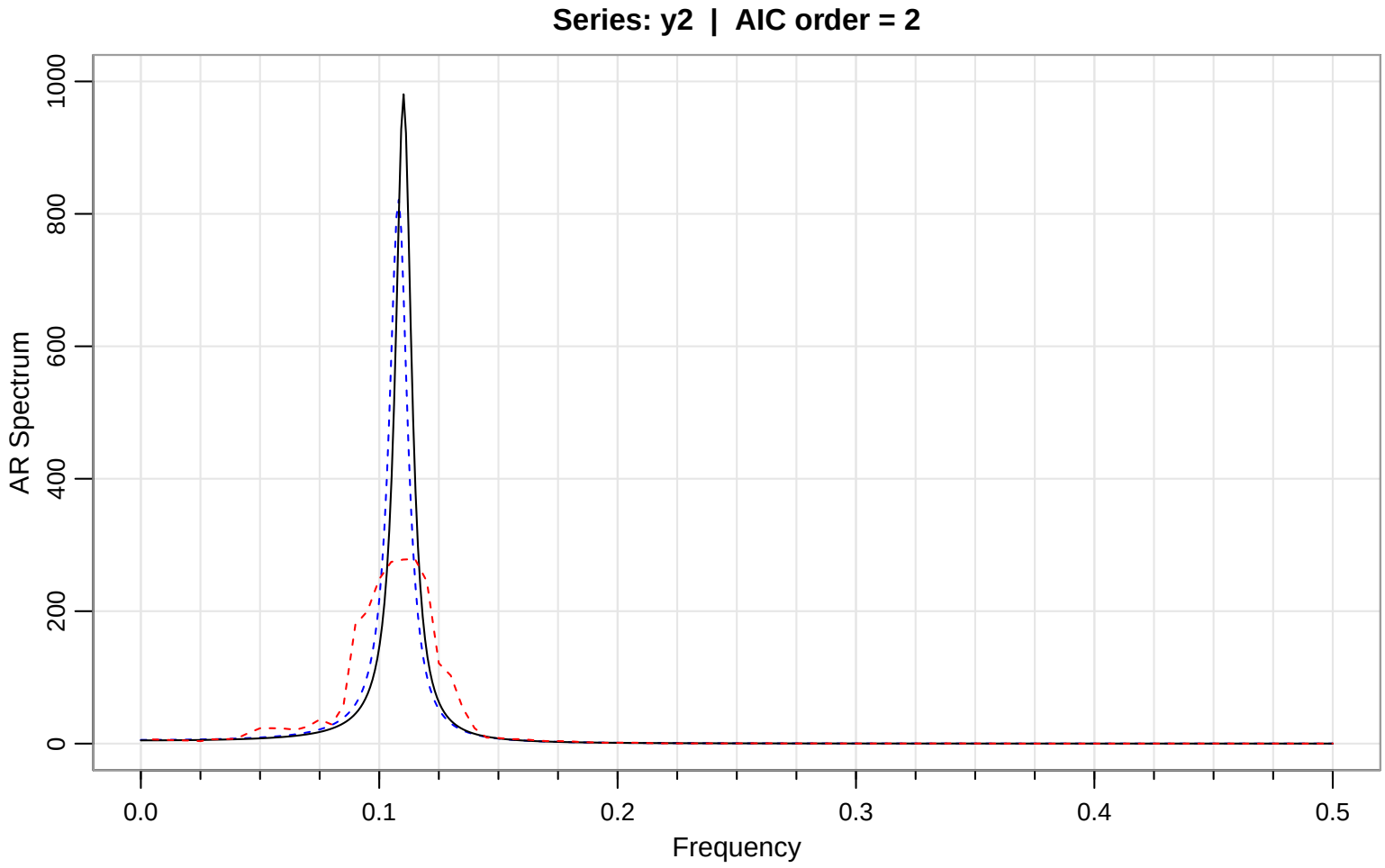
Example: ARMA(1, 1) with $\phi = 0.8$ and $\theta = 0.5$

1. Use AIC to select p, the order of the AR model
2. Use the estimates $(\hat{\phi}_1, \hat{\phi}_2, \cdots, \hat{\phi}_p, \hat{\sigma}^2)$ to compute the estimated spectral



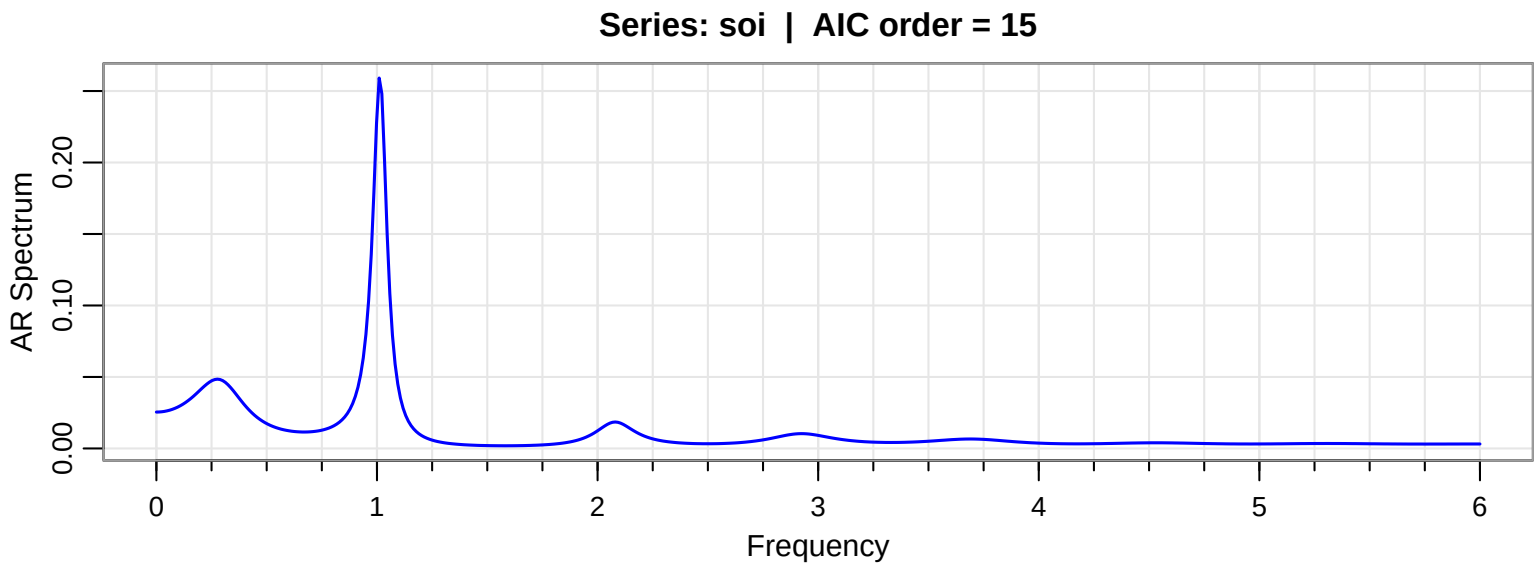


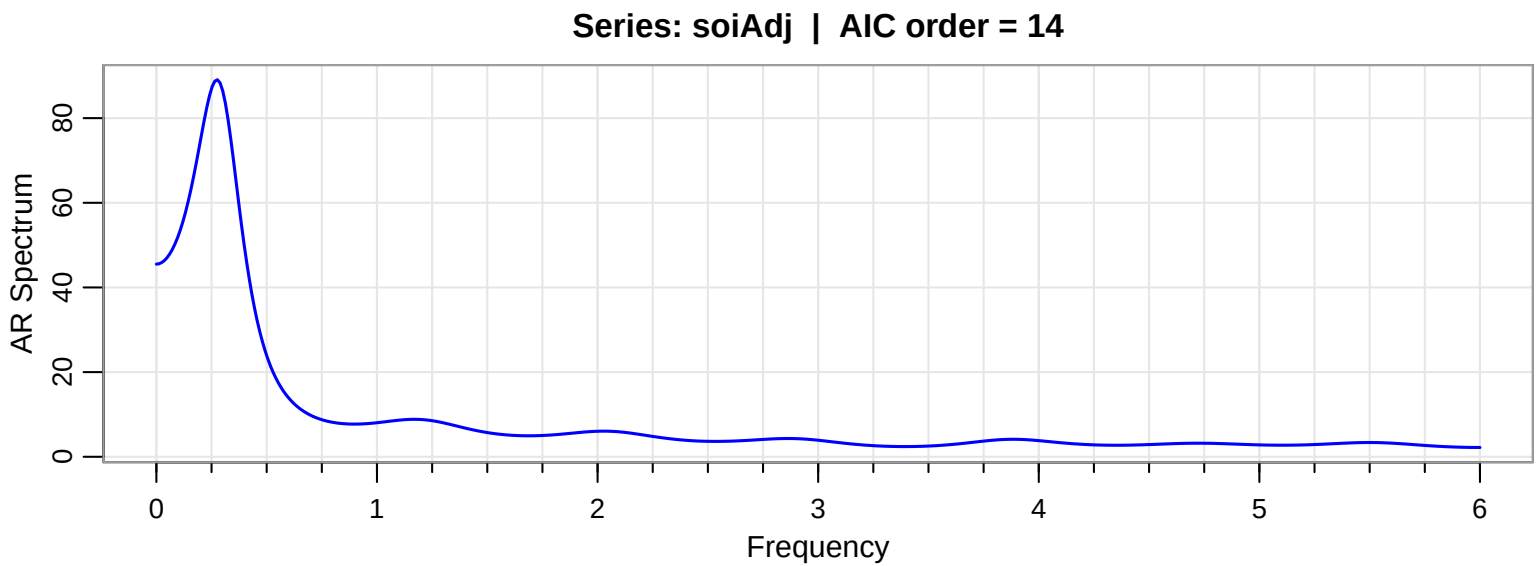
Example: AR(2) with $\phi_1 = 1.5$ and $\phi_2 = -0.95$



- Black: True
- Blue: Parametric Estimate
- Red: Non-parametric Estimate (Daniell $l = 3$)

SOI Example


Series: soi | AIC order = 15
AR Spectrum
0.00
0.10
0.20
0
1
2
3
4
5
6
Frequency

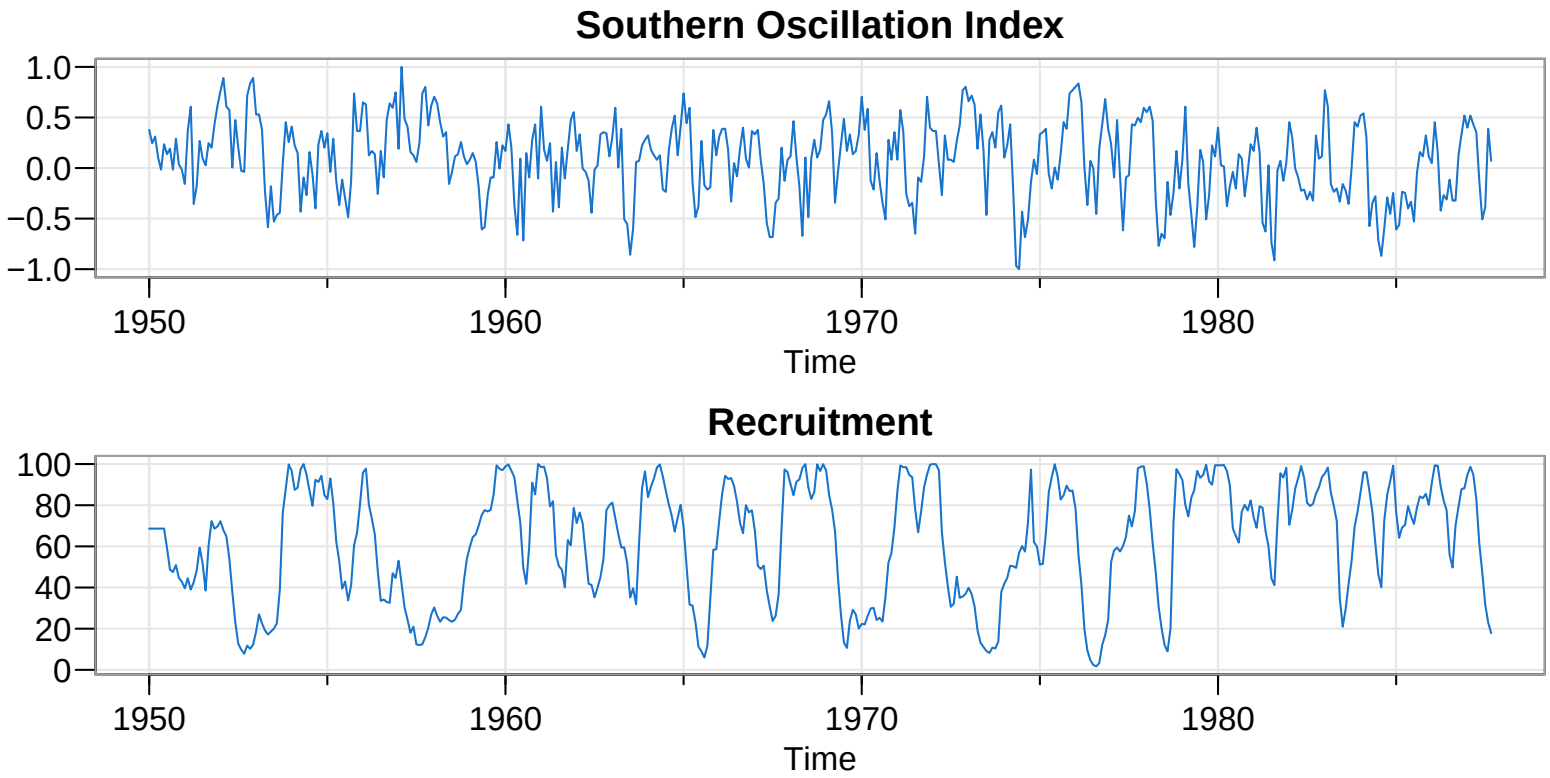

Series: soiAdj | AIC order = 14
AR Spectrum
0
20
40
60
80
0
1
2
3
4
5
6
Frequency

Why Lagged Regression?

$$Y_t - \mu_Y = \sum_{j=0}^{\infty} \beta_j (X_{t-j} - \mu_X) + V_t$$

- X_t: input time series; Y_t: output time series; V_t: error time series
- β_j: response at lag j to a unit input impulse
- **Goal**: identify delay, duration, and strength

Example: SOI and Recruitment



Input: SOI **Output**: recruitment

Question: Does SOI lead recruitment, and by how many months?

Lagged Regression: Two Complementary Views

- Time domain: CCF $\rightarrow$ prewhitening $\rightarrow$ lag response
- Frequency domain: cross-spectrum $\rightarrow$ coherence $\rightarrow$ transfer function

Both describe linear dynamic association; neither alone establishes causality

Cross-Correlation: Fix the Lag Convention

The cross-covariance function of two jointly stationary processes $\{X_t\}$ and $\{Y_t\}$ is

$$\gamma_{XY}(h) = \mathbb{E}\left[(X_{t+h} - \mu_X)(Y_t - \mu_Y)\right].$$

The cross-correlation function (CCF) is

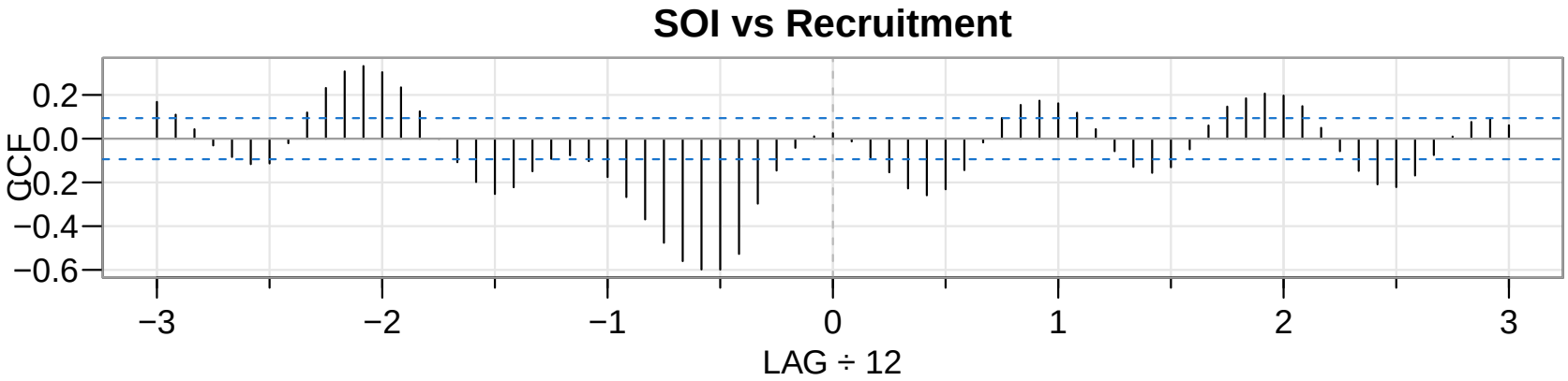
$$\rho_{XY}(h) = \frac{\gamma_{XY}(h)}{\sqrt{\gamma_X(0)\gamma_Y(0)}}.$$

For $Y_t = \beta X_{t-\ell} + V_t$, with V_t uncorrelated with X at all lags,

$$\gamma_{XY}(h) = \beta\gamma_X(h + \ell).$$

Lag convention: A negative CCF lag indicates that X leads Y.

SOI-Recruitment: What Does the CCF Show?



CCF of SOI and recruitment has a negative trough near $h = -6$. Thus, SOI leads recruitment by 6 months.

This is exploratory evidence of a lead, not a definitive causal delay.

Why can the Raw CCF Mislead?

$$Y_t = \beta X_{t-\ell} + V_t \Rightarrow \gamma_{XY}(h) = \beta \gamma_X(h + \ell)$$

- Autocorrelated input $\Rightarrow$ broad CCF peaks
- CCF reflects both input persistence and response delay
- **Prewhitening** isolates the lag structure more clearly

Prewhitening: Filter Both Series

Step 1: Fit an ARMA model to the input X_t:

$$X_t = \frac{\theta_X(B)}{\phi_X(B)} W_t, \qquad \underbrace{\frac{\phi_X(B)}{\theta_X(B)}}_{A(B)} X_t = W_t.$$

Step 2: Apply the same filter to the output Y_t:

$$Y_t = \beta(B) X_t + V_t$$

$$\underbrace{A(B) Y_t}_{\tilde{Y}_t} = \beta(B) W_t + A(B) V_t.$$

Step 3: Examine the CCF of W_t and $\tilde{Y}_t$ to identify potential input-output lags.

Key principle: Estimate the filter from X_t, then apply it to **both series** (after centering)

Prewhitened CCF Reveals the Response

After prewhitening, let

$$\tilde{Y}_t = \sum_{j=0}^{\infty} \beta_j W_{t-j} + \tilde{V}_t,$$

where W_t is white noise with variance σ_W^2.

If $\tilde{V}_t$ is uncorrelated with W_t at all lags,

$$\boxed{\gamma_{W,\tilde{Y}}(-j) = \mathrm{Cov}(W_t, \tilde{Y}_{t+j}) = \sigma_W^2 \beta_j}$$

- **Lag $-j$:** Input W_t precedes the response $\tilde{Y}_{t+j}$ by j
- **CCF magnitude:** Proportional to the response coefficient β_j
- **Normalized CCF:** Reveals the response pattern, not the coefficients directly

Key insight: Prewhitening makes the CCF proportional to the input–output impulse response.

Time-Domain Identification: Workflow

1. **Model the input:** Fit an ARMA model to X_t

2. **Prewhiten:** Apply the same filter to X_t and Y_t; examine their filtered CCF

3. **Identify the response:** Fit a parsimonious transfer-function model with appropriate delays

4. **Validate the model:** Model residual autocorrelation and check residual diagnostics

Parsimonious Transfer Functions

A transfer function summarizes the delayed response of Y_t to changes in X_t:

$$Y_t = \beta(B)X_t + V_t, \qquad \boxed{\beta(B) = B^d \frac{\delta(B)}{\omega(B)}}.$$

- d: **Delay** before the response begins
- $\delta(B)$: **Numerator** controlling the response shape
- $\omega(B)$: **Denominator** controlling persistence and decay

Example:

$$\beta(B) = \frac{0.5B^2}{1 - 0.6B} = 0.5B^2 + 0.3B^3 + 0.18B^4 + \cdots$$

Interpretation: The response begins after 2 periods and then decays geometrically

From Cross-Covariance to Cross-Spectrum

Time domain: The cross-covariance $\gamma_{XY}(h)$ describes dependence across lags

Frequency domain: Its Fourier transform gives the cross-spectral density:

$$\boxed{f_{XY}(\omega) = \sum_{h=-\infty}^{\infty} \gamma_{XY}(h) e^{-2\pi i \omega h}}$$

Unlike an ordinary spectrum, the cross-spectrum is generally **complex-valued**:

$$f_{XY}(\omega) = |f_{XY}(\omega)| e^{i\phi_{XY}(\omega)}, \qquad f_{YX}(\omega) = \overline{f_{XY}(\omega)}.$$

- **Magnitude:** Strength of cross-frequency association.
- **Phase:** Relative timing of frequency components.

Key idea: The cross-spectrum describes how two series are related at different frequencies.

Squared Coherence: Frequency-Specific R^2

The cross-spectrum measures frequency-specific association, but its magnitude depends on the scales of X_t and Y_t.

The squared coherence normalizes the cross-spectrum:

$$C_{XY}(\omega) = \frac{|f_{XY}(\omega)|^2}{f_X(\omega) f_Y(\omega)}, \qquad 0 \leq C_{XY}(\omega) \leq 1.$$

- **Near 1:** Strong linear association at frequency ω.
- **Near 0:** Weak linear association at frequency ω.
- **Analogy:** Squared coherence plays a role similar to R^2 in regression.

Estimating Coherence: Smoothing Is Essential

As in **spectral density estimation**, we smooth periodogram ordinates across nearby frequencies to obtain stable estimates

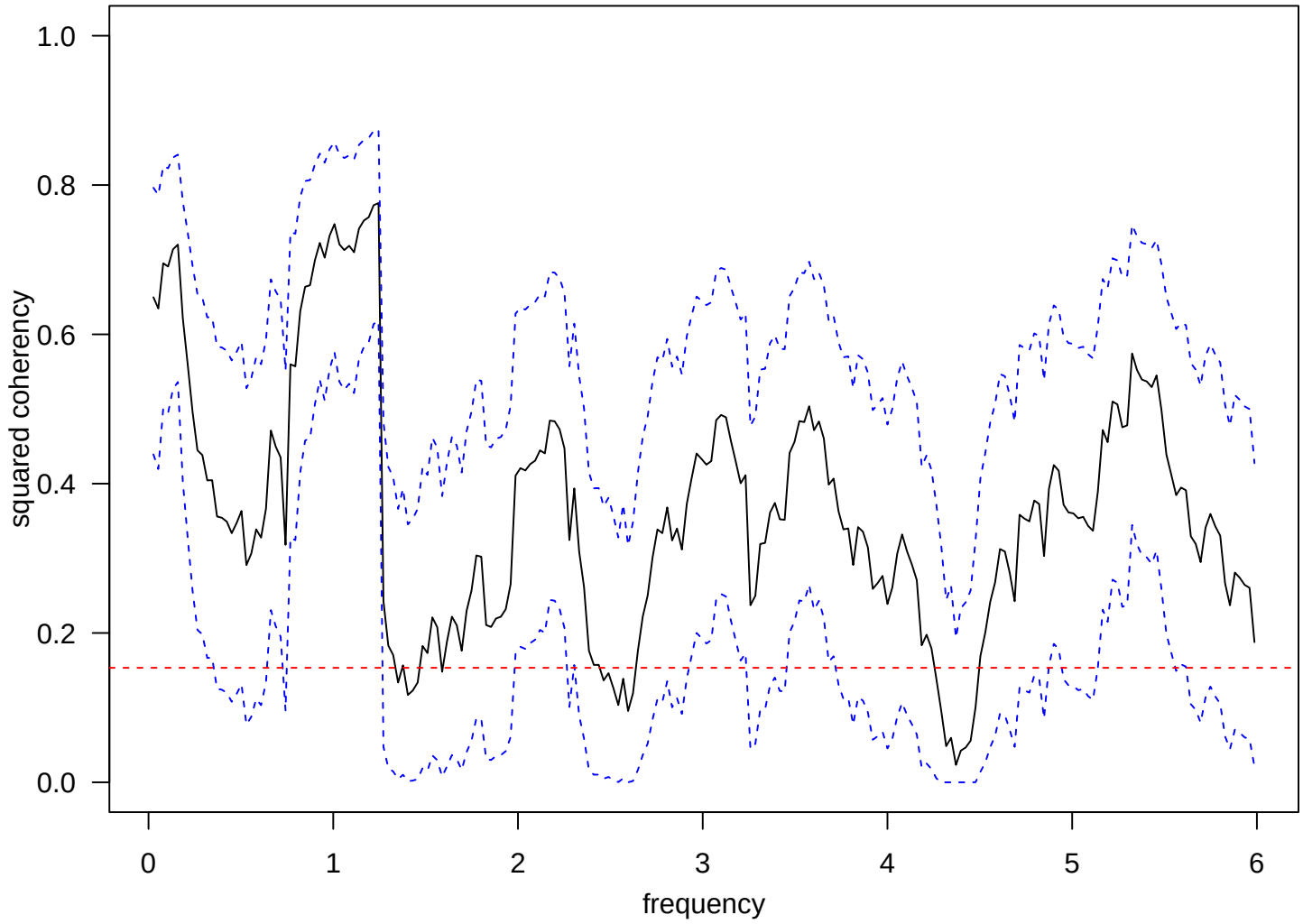
Smoothed cross-spectrum:

$$\hat{f}_{XY}(\omega_j) = \sum_k w_k \, d_X(\omega_{j+k}) \overline{d_Y(\omega_{j+k})}.$$

Estimated squared coherence:

$$\hat{C}_{XY}(\omega_j) = \frac{|\hat{f}_{XY}(\omega_j)|^2}{\hat{f}_X(\omega_j)\hat{f}_Y(\omega_j)}.$$

SOIRecruitment: Squared Coherence



Question: At which frequencies is their linear association strongest?

Two-Sided Filtering vs. Causal Lagged Regression

Unrestricted two-sided linear projection:

$$\widehat{Y}_t = \sum_{j=-\infty}^{\infty} \beta_j X_{t-j}.$$

Its normal equations yield the frequency-domain ratio

$$B(\omega) = \frac{f_{YX}(\omega)}{f_X(\omega)} \quad \text{where } f_X(\omega) > 0.$$

Causal lagged regression: restrict $\beta_j = 0$ for $j < 0$ so the output uses only current and past inputs.

The simple spectral ratio solves the unrestricted two-sided problem; a causal filter generally requires spectral factorization or constrained normal equations.

Transfer Function: From Convolution to Multiplication

Recall the two-sided linear projection:

$$Y_t = \sum_{j=-\infty}^{\infty} \beta_j X_{t-j} + V_t, \qquad \mathrm{Cov}(V_t, X_s) = 0 \quad \forall s.$$

Time domain: Cross-covariance is a convolution:

$$\gamma_{YX}(k) = \sum_{j=-\infty}^{\infty} \beta_j \gamma_X(k-j).$$

Frequency domain: Convolution becomes multiplication:

$$\boxed{f_{YX}(\omega) = H(\omega) f_X(\omega), \qquad H(\omega) = \sum_{j=-\infty}^{\infty} \beta_j e^{-2\pi i \omega j}.}$$

Key insight: The transfer function $H(\omega)$ describes how the input is transformed at each frequency.

Recovering the Impulse Response

From $f_{YX}(\omega) = H(\omega) f_X(\omega)$, we obtain the transfer function:

$$\boxed{H(\omega) = \frac{f_{YX}(\omega)}{f_X(\omega)}, \qquad f_X(\omega) > 0.}$$

The **inverse Fourier transform** recovers the impulse-response coefficients:

$$\boxed{\beta_j = \int_{-1/2}^{1/2} H(\omega) e^{2\pi i \omega_j} \, d\omega, \qquad j \in \mathbb{Z}.}$$

- **Estimate:** Smooth the auto- and cross-spectra; stabilize the spectral ratio.
- **Invert:** Recover the two-sided impulse response $\{\beta_j\}$.

Coherence and Unexplained Variation

Spectral decomposition:

$$f_Y(\omega) = \underbrace{|H(\omega)|^2 f_X(\omega)}_{\text{Explained}} + \underbrace{f_V(\omega)}_{\text{Unexplained}}$$

Using squared coherence,

$$f_V(\omega) = f_Y(\omega)\big[1 - C_{XY}(\omega)\big].$$

The minimum mean squared error is

$$\text{MSE} = \int_{-1/2}^{1/2} f_V(\omega)\, d\omega.$$

Key insight: Squared coherence measures the fraction of spectral variation explained by the optimal two-sided linear projection.

Key Takeaways

1. **Raw CCF can be misleading:** Input autocorrelation may obscure the underlying lag structure.

2. **Prewhitening reveals delays:** Filtering both series helps identify the dynamic input–output response.

3. **Coherence measures association:** Quantifies frequency-specific linear dependence, but not direction or causality.

4. **Spectral methods recover responses:** The spectral ratio identifies the optimal two-sided linear filter, not necessarily the optimal causal predictor.

Bottom line: Time-domain and frequency-domain methods provide complementary views of dynamic relationships between time series.